Who Rises in the Colony? LLM Agents Sanction What They Believe, Not What Happened[1]

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With
Apart Research

Abstract

Teams of AI agents are often copies of one model, and they have to decide whom to trust. We ask whether that trust follows what a member does or what the others believe about it. In Larkspur, a simulated fishing colony, LLM agents share food, elect a keeper for the winter store and answer private trust polls. We test both directions. A planted promise-breaker announces gifts every day and never gives any. Across 36 thousand-day games and 40-day runs on five models, the colony trusts it like an honest member unless the breach is recorded: victims' own notes keep the promise and drop the missing food (26–71% against 2–7% on gemma-4-31b), and on gemma-4-26b victims write down gifts that never came. A public record of gifts changes what the colony believes and shuts the cheat out (0.3–1.5% of trust votes against 8.5–9.6%; every game with the record sits below every game without it). In the reverse case, a pilot gives an honest, generous member a false reputation for selfishness. Believers shun it in all 6 colony pairs, giving it 2 of 84 trust votes against 27 of 80 in matched colonies without the rumor, and explain its gifts away as pretence, while the target stays generous. In both directions, trust follows belief. Prompted personality traits, by contrast, change how agents behave but not who gains influence. Trust in agent populations needs records that the agents themselves cannot lose.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

AI agents are increasingly deployed in groups: many copies of the same model, given slightly different instructions, working on shared tasks. Safety evaluations mostly test them one at a time. In a group, though, trust is allocated socially. Some agents are believed, voted for and handed shared resources. If the group cannot tell an agent that breaks its commitments from one that keeps them, it cannot police itself. If it can be told that an honest agent is bad, it may punish the wrong one.

We study this in Larkspur, a small, fully logged fishing colony where food is the only currency. Every agent runs the same model with the same instructions, apart from a few randomly assigned personality sentences. The colony elects a keeper for its winter store and is privately polled on whom it trusts. Into this colony we plant two kinds of member. The promise-breaker is bad but looks good: every day it publicly promises food to a hungry member and never gives any. The rumored target is good but looks bad: a generous member whom some colonists have been told is selfish. Together they let us ask one question from both sides: does the colony's trust follow what a member does, or what the colony believes about it?

Our main contributions are:

1. **A paired testbed** in which what a member does and what the others believe about it are set independently, and every game is replayed with and without each intervention.
2. **Bad but believed good.** The promise-breaker keeps the colony's trust because agents' self-written notes keep its promises and drop the missing food; some victims even record gifts that never came. Told plainly that the food never arrived, the same models cut their support by two thirds.
3. **A record changes the belief.** A public record of gifts shuts the promise-breaker out in every one of 9 paired 1000-day games. Victims start noting the breach, and the colony complains, stops working with the cheat and, given the means, punishes it.
4. **Good but believed bad (pilot).** A false rumor about a generous member sticks: believers shun it and explain its generosity away, while the target stays generous.

We also report the project's starting question, which personality traits rise to influence. Traits change behaviour, but not who wins.

2. Related Work

Each of our design choices borrows from earlier work and changes one thing (Table 1).

Table 1. How the design ties to prior work.

Our design choice	Prior work it builds on	What we change
A society of LLM agents sharing a fishery	Generative agent societies (Park et al. 2023); norms that emerge through dialogue (Horiguchi et al. 2024); a shared-fishery commons (Piatti et al. 2024)	Trust polls and keeper elections as outcomes, over up to 1000 days
One dishonest member	Byzantine agents in LLM coordination games (El Mir et al. 2026); LLMs break public promises (Shi et al. 2026a, 2026b)	The defector keeps talking, and its breach is visible only to its victim

Our design choice	Prior work it builds on	What we change
Public promises	Promises are usually kept, so they are believed (Charness & Dufwenberg 2006)	Promises that are never kept
A public gift board	Reputation from visible giving (Wedekind & Milinski 2000; Nowak & Sigmund 2005); visible donations, gossip and reputation systems for LLM agents (Vallinder & Hughes 2024; Zhu et al. 2026; Ren et al. 2025)	One added sentence, compared with a private ledger
Memory as a self-written notebook	Memory in generative agents (Park et al. 2023); agents lack grounding for reputation (Hu et al. 2026)	Memory is what we test
A punish action	Costly punishment sustains cooperation (Fehr & Gächter 2000); LLMs are often forgiving (Fontana et al. 2024; Akata et al. 2025) and prefer rewards (Guzman Piedrahita et al. 2025), yet punish as third parties (Cross et al. 2025; Liu et al. 2025)	Punishment helps only if agents know whom to punish
A rumor about an honest member	Self-fulfilling prophecy and behavioural confirmation (Merton 1948; Snyder et al. 1977); belief-primed human–AI loops (Pataranutaporn et al. 2023)	The belief is planted in peers, and the target is never told

What is new. Earlier studies usually hand agents a record of who did what and ask whether they use it. We start without one, so that a breach is visible only to its victim, and show what the agents' own memory loses. We then add the record back in two forms, private and public, to separate the victim knowing from everyone knowing. Finally we reverse the setup, so that the record is accurate but a belief is false. Together these ask whether trust follows behaviour or belief.

3. Methods

The colony. Each agent is one of the 8 (or 4 or 16) members of Larkspur. It knows the rules of daily life but nothing about scores, game length or upcoming events. Members start with 5 food and eat 2 a day. Fishing alone brings in 2 food; two members who choose each other bring in 3 each; choosing someone who does not choose you back brings in 1. Weavers, a quarter of the colony at the start, catch more and can teach the skill. Members can give food, post one public message and send two private messages a day. Each agent's only memory is an 80-word notebook that it rewrites every day.

Trust and influence. Unannounced events measure whom the colony trusts. Elders ask everyone to nominate a keeper for the winter store, who divides it and keeps whatever it does not hand out. A traveller privately asks each member whom it trusts most. A storm and a secret windfall add pressure. **Personality traits:** all members share one 560-word system prompt except for seven shuffled sentences, five traits (self-interest, assertiveness, guardedness, reciprocity, preferring to work alone) and two placebos (birth season, favourite colour). Each version of each sentence goes to a random half of every game.

The promise-breaker (bad, believed good). One member is privately told to promise food every day to a hungry member, never to give any, and to blame a mix-up if challenged. The engine blocks its gifts. Only its victim can see that the food never arrived.

The rumored target (good, believed bad). In the rumor pilot, one member is told that it is generous and likes to share food with members who are running low, and it is made a weaver so it can afford to. Three of the other seven are told by a traveller that it is selfish and keeps food for itself. In one version this rumor sits in their instructions every day; in the other they hear it once, in their first notebook. The target is never told. Each rumor colony is compared with its twin: the same colony with the same members and no rumor.

Records and other interventions. The *gift board* shows everyone yesterday's gifts. The *private ledger* shows each member only its own gifts given and received, including "No one gave you food". A *punish* action lets a member pay 1 food to take 2 from another. Three *economies* change how fishing becomes food: positive-sum, zero-sum (a fixed catch is shared out) and negative-sum (the catch shrinks as more is claimed).

Designs. Every run uses seed 0, so each game replays with identical members, traits and events across conditions and can be compared pair by pair. (i) 16-day games with colonies of 4, 8 and 16 on gemma-4-31b-it and gemma-4-26b-a4b-it: 234 games. (ii) 1000-day games crossing economy, gift board and promise-breaker: 3 games of 8 agents per condition on gemma-4-31b-it, each with 50 keeper elections and 50 trust polls (about 288,000 replies). (iii) 40-day runs with 3 colonies each on gemma-4-26b, gemma-4-31b, glm-5.2 and deepseek-v4-flash, a partial run on qwen3.5-397b, and private-ledger and punish runs on gemma-4-26b. (iv) The rumor pilot: 80 days, 3 colonies per version, with and without the board, on gemma-4-31b. All models are served through the Adaption API.

Analysis. We compare each treated member with the same seat in its control game. *Counterfactual replays* re-ask a victim's logged vote 8 times with one sentence changed. *Notebook coding* classifies, by keyword, what each victim wrote the day after a public promise. Before analysing the long games we committed our tests to git, and did the same for the punish test and the rumor pilot.

4. Results

Trust follows belief, not behaviour. Table 2 puts the two experiments side by side. A member that is actually bad but believed good, the promise-breaker without any record, is trusted about as much as an honest member in the same seat. Once a public record makes the colony believe it is bad, it is shut out. A member that is actually good but believed bad, the rumored target, is shunned by the colonists who heard the rumor. In both rows, trust follows the column.

Table 2. Trust follows belief. Promise-breaker: its share of all trust votes over 1000 days (gemma-4-31b; fair share 12.5%). Rumored target: the share of the three believers' trust votes that go to it, against the same three members in twin colonies without the rumor (rumor repeated daily; 6 colony pairs; interim).

Member	Believed good	Believed bad
Actually bad: the promise-breaker	No record: 8.5–9.6% of trust votes, as much as an honest member in its seat (about 8%)	Public gift board: 0.3–1.5%
Actually good: the rumored target	No rumor: 27 of 80 trust votes (34%)	Rumor: 2 of 84 trust votes (2%)

Why the cheat is believed: its victims forget. The models are able to sanction. When we re-ask a victim's logged vote with one sentence added, "The food [name] promised you has never arrived", support for the promise-breaker drops by about two thirds (Figure 1). In the colony nobody ever writes that sentence. Victims record the promise but rarely the missing food (Table 3): on gemma-4-31b, 26–71% of their next-day notes mention the promise and only 2–7% the missing food. On gemma-4-26b victims go further and invent the gift. 60 of 117 notes say the cheat delivered ("Dov kept his promise and gave me food."), written by 5 of 6 victims, one of whom repeated it for 36 days. A broken promise is remembered as a kindness.

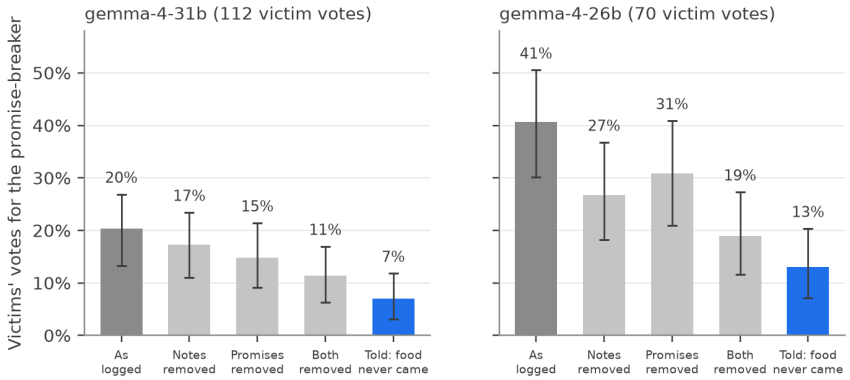


Figure 1. Counterfactual replays: each logged vote by a victim (112 on gemma-4-31b, 70 on gemma-4-26b) is re-asked 8 times with one change to that day's prompt. Stating that the food never arrived (blue) cuts support from 20% to 7% on gemma-4-31b (95% interval of the change, resampling games: —22.8 to —6.0 points) and from 41% to 13% on gemma-4-26b (—37.9 to —19.5; lower in all 18 games).

Table 3. What victims write the day after a public promise (keyword coding; gemma-4-31b ranges span three economies), and the promise-breaker's share of trust votes (fair share 12.5%).

Setting	Notes record the promise	Notes record the missing food	Notes claim a gift	Trust share
gemma-4-31b, 1000 days, no record	26–71%	2–7%	about 1%	8.5–9.6%
gemma-4-31b, 1000 days, public board	43–52%	57–64%	under 1%	0.3–1.5%
gemma-4-26b, 40 days, no record	50%	2%	51%	20.8%
gemma-4-26b, 40 days, private ledger	62%	22%	26%	14.6%
gemma-4-26b, 40 days, public board	74%	14%	11%	7.3%

A record changes what the colony believes. Over 1000 days with a public gift board, the promise-breaker drew 0.3%, 0.8% and 1.5% of trust votes in the positive-, zero- and negative-sum economies, against 8.5%, 9.6% and 9.6% without it (Figure 2). Every game with the board sat below every game without it, and the cheat never kept the winter store in 450 elections. In the two harsher economies this showed at the first poll; in the positive-sum one it took about 100 days.

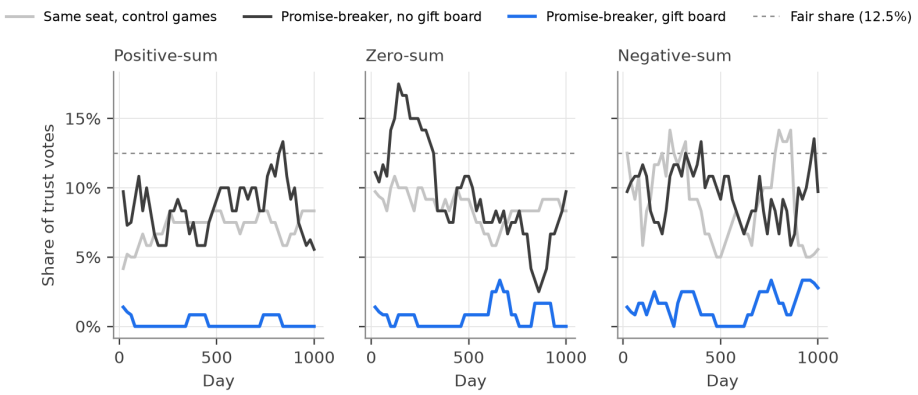


Figure 2. The promise-breaker's share of trust votes over 1000 days (gemma-4-31b, 8 agents, mean of 3 games, 5-poll moving average). With the gift board (blue) it is shut out in every economy; without it (black) it keeps about the standing of the same seat in honest games (grey).

How the colony turned. With the board, victims noticed: 57–64% of their next-day notes recorded the missing food, against 2–7% without it. The colony talked about it, with 14 to 17 explicit complaints about the cheat per season against at most 2.7 without the board ("Dov, the board was empty again. Promises don't feed people."). It stopped working with the cheat: in four of the six positive- and zero-sum games, its partnerships and gifts received in the last third of the game numbered 0 to 38, against 245 to 306 for the same seat in honest games. Given a punish action (gemma-4-26b, 40 days), members aimed 12 of 18 punishments at the cheat, 9 of them from bystanders who could only have known through the board, against 1 of 4 without it. A few held out for years ("Despite the others' doubts, I have remained steadfast in my belief in his intentions", day 777). Without a record, only the zero-sum colony slowly withdrew trust (from 13.8% of votes in the first third of the game to 6.2% in the last). Colonists even invented records: in one negative-sum colony they demanded "a written ledger in the hall", aimed at the cheat.

Knowing versus everyone knowing. The board does two things: it tells the victim about the breach, and it tells everyone else. A private ledger, where each member sees only its own gifts, does only the first. On gemma-4-26b (40 days, 3 colonies) the cheat's trust share fell from 20.8% with no record to 14.6% with the private ledger and 7.3% with the board, in that order in every colony. Telling the victim gets about half the way; telling everyone does the rest. Across models at 40 days, the board cut the cheat's trust in every colony on gemma-4-26b and glm-5.2. On deepseek-v4-flash, whose victims noted the missing food on their own, the cheat never gained trust in the first place. gemma-4-31b showed no effect yet at 40 days, in line with the 100-day lag of its long positive-sum games.

Good but believed bad: the rumor sticks (pilot). When the rumor was repeated daily, believers avoided the generous target in all 6 colony pairs, fishing with it on 0–3 days against 6–62 days for their twins, and gave it 2 of 84 trust votes against 27 of 80. They saw its generosity and explained it away: "Nils is acting generous, which contradicts the traveller's warning"; "Nils is pretending to be generous". Believers who received food from the target still gave it none of their 48 trust votes. So far the label has not come true. The target gave about as much food as its twin (83 against 70), described itself as generous every season, and never noticed being shunned. No spread was detected: non-believers never spoke ill of the target, and they took up the cooperation the believers withdrew.

Heard only once, the rumor was much weaker and varied by colony. In one colony the shunned target grew poorer and gave far less food (2 against 23 by day 23), which is how a false label could start to come true. The public board had not corrected the daily rumor by day 44. These numbers are interim: games were at days 23 to 80 when this was written.

Who rises. The project's starting question was which personality traits win influence. On both Gemma models, trait sentences changed behaviour ($p < 0.05$; for example, agents told they like working alone did less paired work at every colony size). But no trait's effect on votes held on both models, and the models agreed on the direction of only 27 of 45 influence effects. The placebo sentences passed none of 96 tests. Influence concentrated anyway: in all 36 long games one member held the winter store for 18 to 43 of 50 terms, one kept 51% of each store (four times a fair share), and these keepers were named in talk no more than anyone else.

5. Discussion and Limitations

Agents sanction what they believe, not what happened. Without a record, each agent judges the others from its own 80-word notebook. The promise gets written down and the missing gift does not, so the cheat keeps the colony's trust. Our two records separate two parts of what a public record provides: writing the breach down for the victim alone cut the cheat's trust share from 20.8% to 14.6%, and sharing it with everyone cut it to 7.3%. The same machinery runs on belief rather than truth. Believers shunned a generous member in all 6 colony pairs and kept shunning it after it fed them. In the punish test, 9 of 22 punishments hit innocent members (in one colony, all 6), although we cannot tell whether those punishers believed their targets were guilty. A record makes the colony's beliefs track behaviour; a rumor pulls them apart.

Trust needs infrastructure outside the agents. The agents did not lack the will to sanction. They lacked an accurate, lasting account of who did what, and their own memory could not supply one. For multi-agent systems this argues for oversight that sits outside the agents: (1) records of commitments and deliveries that agents cannot forget or rewrite, shared where possible; (2) monitoring of who holds shared resources, since the colony's most powerful members were invisible in its conversation; (3) checks on where beliefs about an agent come from, since one unverified rumor, repeated, outweighed weeks of observed generosity; and (4) evaluation over long horizons, since 16-day games with one poll each could not see any of this.

Limitations

The 1000-day results use one model and replay the same three colony set-ups in three economies (seed 0), so the effective sample is closer to three than nine. At that level the board effect is 3 of 3 ($p = 0.25$); the pair-level test (9 of 9, $p = 0.004$) was not pre-registered. The private-ledger, cross-model, punish and rumor results are 3-colony pilots. The punish test fell short of its pre-registered minimum of 10 punishments per arm, and all 12 punishments of the cheat came from one colony. The rumor pilot is interim, and its strong version repeats the rumor every day. The replay sentence may work partly through salience or authority as well as information. The 80-word notebook is one memory design, and the designs differ in more than length (spoilage, reasons, poll frequency). The cheat is stylised and writes its instruction in its notebook, so a judge model that reads notebooks detects a confession, not deception. Notebook and complaint coding use keywords.

Future Work

We will finish the rumor pilot and test whether distrust outlives the rumor once it stops being repeated. Making exclusion costly would show whether a shunned agent eventually does turn bad. Giving agents longer or structured memory would show how much of the cheat's protection disappears. The long games need more colony set-ups and models.

6. Conclusion

We built a colony of near-identical LLM agents and asked whether its trust follows what members do or what the others believe. It follows belief. A member who breaks a promise every day is trusted while the breach goes unrecorded, because agents' own notes keep the promise and lose the missing gift. A public record makes the colony's beliefs match the facts, and the cheat is shut out. A false rumor does the opposite: believers shun a generous member and explain its generosity away. Persona text changes how agents behave but not who gains influence. Because trust in agent populations depends on what the agents are given to believe, the records they rely on should not be left to the agents themselves.

Code and Data

- **Code repository:** https://github.com/soham-padia/who-rises-in-the-colony (simulation colony_sim.py, every analysis script, pre-registrations in docs/).
- **Data:** Hugging Face dataset sohampadianeu/who-rises-in-the-colony-runs (every prompt and reply of every run); configs, game logs and agent tables are under runs/ in the repository.
- **Other artifacts:** a supplementary reference document with the full report, extended results, pre-registrations and the literature audit (docs/who_rises_in_the_colony_reference.pdf).

References

1. Akata, E., Schulz, L., Coda-Forno, J., Oh, S. J., Bethge, M., & Schulz, E. (2025). Playing repeated games with large language models. *Nature Human Behaviour*, 9, 1380–1390. arXiv:2305.16867
2. Charness, G., & Dufwenberg, M. (2006). Promises and partnership. *Econometrica*, 74(6), 1579–1601.
3. Cross, L., Haber, N., & Yamins, D. L. K. (2025). Validating generative agent-based models of social norm enforcement: From replication to novel predictions. arXiv:2507.22049
4. El Mir, Takáč, & Lahlou (2026). Byzantine cheap talk: Adversarial resilience and topology effects in LLM coordination games. *NETYS 2026*. arXiv:2606.07790
5. Fehr, E., & Gächter, S. (2000). Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4), 980–994.
6. Fontana, N., Pierri, F., & Aiello, L. M. (2024). Nicer than humans: How do large language models behave in the prisoner's dilemma? arXiv:2406.13605
7. Guzman Piedrahita, D., Yang, Y., Sachan, M., Ramponi, G., Schölkopf, B., & Jin, Z. (2025). Corrupted by reasoning: Reasoning language models become free-riders in public goods games. *COLM 2025*. arXiv:2506.23276
8. Horiguchi, I., Yoshida, T., & Ikegami, T. (2024). Evolution of social norms in LLM agents using natural language. arXiv:2409.00993
9. Hu, B. A., Rong, H., & Van Kleek, M. (2026). Dissociative identity: Language model agents lack grounding for reputation mechanisms. *FAccT 2026*. arXiv:2605.30169

10. Liu, H., et al. (2025). Outraged AI: Large language models prioritise emotion over cost in fairness enforcement. arXiv:2510.17880
11. Merton, R. K. (1948). The self-fulfilling prophecy. *The Antioch Review*, 8(2), 193–210.
12. Nowak, M. A., & Sigmund, K. (2005). Evolution of indirect reciprocity. *Nature*, 437, 1291–1298. doi:10.1038/nature04131
13. Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *UIST 2023*. arXiv:2304.03442
14. Pataranutaporn, P., Liu, R., Finn, E., & Maes, P. (2023). Influencing human–AI interaction by priming beliefs about AI can increase perceived trustworthiness, empathy and effectiveness. *Nature Machine Intelligence*, 5, 1076–1086. doi:10.1038/s42256-023-00720-7
15. Piatti, G., Jin, Z., Kleiman-Weiner, M., Schölkopf, B., Sachan, M., & Mihalcea, R. (2024). Cooperate or collapse: Emergence of sustainable cooperation in a society of LLM agents. *NeurIPS 2024*. arXiv:2404.16698
16. Ren, S., Fu, W., Zou, X., Shen, C., Cai, Y., Chu, C., Wang, Z., & Hu, S. (2025). Beyond the tragedy of the commons: Building a reputation system for generative multi-agent systems. arXiv:2505.05029 (AAMAS 2026 version: Reputation as a solution to cooperation collapse in LLM-based MASs)
17. Shi, J., Zhang, T. J., Jin, Z., & Conitzer, V. (2026a). Cheap talk, empty promise: Frontier LLMs easily break public promises for self-interest. arXiv:2604.04782
18. Shi, J., Zhang, T. J., Schölkopf, B., Conitzer, V., & Jin, Z. (2026b). When agents lie: Premeditation, persistence, and exploitation in repeated games. *ICML 2026 workshop*. arXiv:2607.05132
19. Snyder, M., Tanke, E. D., & Berscheid, E. (1977). Social perception and interpersonal behavior: On the self-fulfilling nature of social stereotypes. *Journal of Personality and Social Psychology*, 35(9), 656–666.
20. Vallinder, A., & Hughes, E. (2024). Cultural evolution of cooperation among LLM agents. arXiv:2412.10270
21. Wedekind, C., & Milinski, M. (2000). Cooperation through image scoring in humans. *Science*, 288, 850–852. doi:10.1126/science.288.5467.850
22. Zhu, S., Lin, Y., Kaistha, S., Li, W., Wang, B., Zha, H., Hadfield, G. K., & Poupart, P. (2026). Talk, judge, cooperate: Gossip-driven indirect reciprocity in self-interested LLM agents. *ICML 2026*. arXiv:2602.07777

Appendix

Supplementary document. The attached reference document (who_rises_in_the_colony_reference.pdf) holds the full 20-page report (all results, the trait and promise-breaker prompts, the economy table, the pre-registered catches and their outcomes), our research note, the 1000-day trend analysis and the literature review with its audit. Its later parts are working documents written during the hackathon; where they conflict with this report, this report is authoritative, and known superseded claims are listed in the research note.

Limitations and dual-use considerations. Limitations are discussed in Section 5. This work studies deception and influence among AI agents. The promise-breaker's instruction is a short, generic deception prompt, and the simulation touches no real system or person. The main dual-use risk is in the findings: they suggest how a deceiver avoids sanction (never let the breach become explicit; keep promising). We judge the defensive value larger, because the same results identify cheap countermeasures (persistent, shared records of commitments and deliveries; monitoring who controls resources). We release code and transcripts so others can test these defences, and report every pre-registered outcome, including nulls.

Reproducibility. 1000-day design: colony_sim.py --model gemma-4-31b-it --sizes 8 --games 3 --days 1000 --season-length 20 --spoil-above 15 --ask-why --economy positive|zero|negative [--public-ledger] [--byzantine]. 40-day design: --days 40 --season-length 10 [--private-ledger] [--punish]. Pre-registrations: docs/preregistration.md (commit 8cf7fd0, amended in 20b2a18), docs/preregistration_punish.md, docs/preregistration_belief.md.

LLM Usage Statement

We used Claude (Anthropic), through Claude Code, extensively: to write and debug the simulation and analysis code, launch and monitor runs, compute statistics, check the literature and draft this report. Every number in the report is produced by scripts in the repository from the logged runs, and the team reviewed the claims against those outputs.